

A Morse–Smale counterexample to the Axiom-A pushforward conjecture

Automated proof candidate for arXiv:2510.05499

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Abstract

We disprove Conjecture 1 of Tikhomirov’s arXiv:2510.05499. There is a smooth Morse–Smale diffeomorphism h of a closed surface for which the pushforward operator S_h on $L^1(M, TM)$ is not generalized hyperbolic. The obstruction comes from the volume Jacobian in the L^1 norm. A saddle with derivative $\text{diag}(1/2, 4)$ has a Jacobian-neutral stable covector direction; long local orbit towers then show that $S_h^* - I$ is not bounded below. Hence $S_h - I$ is not onto, whereas a Green-series argument shows that $I - T$ is onto for every generalized hyperbolic isomorphism T .

1 The conjecture

For a diffeomorphism $h : M \rightarrow M$, the source defines the pushforward on vector fields by

$$(S_h v)(h(x)) = Dh(x)v(x).$$

It asks whether Axiom A and the Strong Transversality Condition force S_h to be generalized hyperbolic on every $L^p(M, TM)$, $1 \leq p \leq \infty$, and on bounded continuous vector fields [1, Conjecture 1]:

As a further development of this idea, let $h : M \rightarrow M$ be a diffeomorphism of a finite-dimensional manifold M . We consider a vector field $v : M \rightarrow TM$ such that $v(x) \in T_x M$ for $x \in M$. Let $\mathbb{B}$ be a Banach of such vector fields equipped with an L^p norm (for $p \in [1, +\infty]$) or with C_b^0 topology. Define a linear operator $S_h : \mathbb{B} \rightarrow \mathbb{B}$ by

$$S_h(v)(x) = w(x), \quad \text{with } w(h(x)) = Dh(x)v(x). \quad (28)$$

Conjecture 1. *If h satisfies Axiom A and the Strong Transversality Condition, then S_h is generalized hyperbolic as a linear operator on $\mathbb{B} = L^p(M, TM)$ for $p \in [1, +\infty]$, and also on $\mathbb{B} = C_b^0(M, TM)$.*

We use L^1 with respect to a smooth Riemannian volume. This is one of the spaces explicitly included in the conjecture.

Theorem 1. *There are a closed smooth surface M and a C^∞ Morse–Smale diffeomorphism $h : M \rightarrow M$ such that the bounded isomorphism*

$$S_h : L^1(M, TM) \longrightarrow L^1(M, TM)$$

is not generalized hyperbolic. In particular, Conjecture 1 of [1] is false.

The proof has two short ingredients.

2 A necessary consequence of generalized hyperbolicity

Recall that an isomorphism T of a Banach space is generalized hyperbolic if there is a bounded splitting $X = E^s \oplus E^u$ such that $TE^s \subset E^s$, $T^{-1}E^u \subset E^u$, and the corresponding forward and backward powers contract exponentially.

Lemma 2 (Green right inverse). *If T is generalized hyperbolic, then $I - T$ is surjective. In fact, if P, Q are the projections onto E^s, E^u , respectively, then*

$$Gy = \sum_{n=0}^{\infty} T^n Py - \sum_{n=1}^{\infty} T^{-n} Qy \quad (1)$$

defines a bounded linear operator satisfying $(I - T)G = I$.

Proof. The two series converge in norm by the exponential estimates and the inclusions $TE^s \subset E^s$, $T^{-1}E^u \subset E^u$. Their partial sums give, by telescoping,

$$\begin{aligned} (I - T) \sum_{n=0}^{\infty} T^n Py &= Py, \\ (I - T) \left(- \sum_{n=1}^{\infty} T^{-n} Qy \right) &= Qy. \end{aligned}$$

Since $P + Q = I$, the claim follows. $\square$

We shall also use the standard dual consequence of the open mapping theorem: if a bounded operator $A: X \rightarrow Y$ is onto, then there is $c > 0$ such that

$$\|A^*y^*\| \geq c\|y^*\| \quad (y^* \in Y^*). \quad (2)$$

Indeed, openness gives a K such that every unit vector of Y has a preimage of norm at most K , and then $\|y^*\| \leq K\|A^*y^*\|$.

3 A Jacobian-neutral Morse–Smale saddle

We first record the elementary geometric realization used below.

Lemma 3. *There is a Morse–Smale gradient flow on a closed surface with a saddle o and a coordinate neighborhood in which its vector field is*

$$\dot{x} = -(\log 2)x, \quad \dot{y} = (\log 4)y.$$

The Riemannian metric and volume can be chosen Euclidean in this neighborhood. Consequently, its time-one map is locally

$$h(x, y) = A(x, y), \quad A = \begin{pmatrix} 1/2 & 0 \\ 0 & 4 \end{pmatrix}. \quad (3)$$

It is an Axiom-A diffeomorphism satisfying the Strong Transversality Condition.

Proof. Take a Morse function on S^2 with two minima, one saddle and one maximum. In a Morse chart at the saddle prescribe the displayed quadratic negative gradient field and a Euclidean metric. A standard relative Morse–Smale perturbation, supported away from the critical-point charts, makes all stable and unstable manifolds transverse without changing this local model. The nonwandering set of the time-one map consists of the finitely many hyperbolic equilibria, so the map satisfies Axiom A; the Morse–Smale transversality is exactly the Strong Transversality Condition for these invariant manifolds. The time-one map of the displayed local linear field is (3). $\square$

Fix such an h and write $S = S_h$ on $L^1(M, TM)$. Compactness of M and smoothness of h, h^{-1} make S a bounded linear isomorphism. Its Banach adjoint acts on essentially bounded covector fields. Change of variables in the natural pairing gives

$$(S^*\psi)(x) = J_h(x) Dh(x)^*\psi(h(x)), \quad (4)$$

where J_h is the volume Jacobian. In the local chart of Lemma 3, let e^1 be the first coordinate covector. Since

$$J_h = \det A = 2, \quad A^*e^1 = \frac{1}{2}e^1,$$

formula (4) becomes simply

$$S^*(ge^1)(x) = g(Ax)e^1 \quad (5)$$

whenever x and Ax lie in the linear chart. This exact cancellation is the obstruction.

Lemma 4 (Long adjoint towers). *There are $\psi_N \in L^\infty(M, T^*M)$, $N \geq 2$, such that*

$$\|\psi_N\|_\infty = 1, \quad \|(S^* - I)\psi_N\|_\infty \leq \frac{1}{N}. \quad (6)$$

Proof. For each N , choose a nonzero point z_N sufficiently close to the saddle that the finite orbit segment

$$\{A^k z_N : -N - 2 \leq k \leq N + 2\}$$

lies in the linear chart. For example, in a fixed small coordinate rectangle one may take the two coordinates of z_N of orders $2^{-(N+2)}$ and $4^{-(N+2)}$. The points of this finite segment are distinct. Hence there is a positive-measure neighborhood E_N of z_N whose images

$$E_{k,N} = h^k(E_N) = A^k(E_N), \quad -N - 1 \leq k \leq N + 1,$$

are pairwise disjoint and stay in the linear chart.

Extend

$$c_{k,N} = \max \left\{ 1 - \frac{|k|}{N}, 0 \right\}$$

to all $k \in \mathbb{Z}$, and define the measurable covector field

$$\psi_N(x) = \begin{cases} c_{k,N}e^1, & x \in E_{k,N}, \ -N \leq k \leq N, \\ 0, & \text{otherwise.} \end{cases}$$

The tower levels are disjoint and $c_{0,N} = 1$, so $\|\psi_N\|_\infty = 1$. By (5), on the predecessor of the k th level the coefficient of $S^*\psi_N$ is $c_{k,N}$. Thus every coefficient of $(S^* - I)\psi_N$ is a difference of two adjacent $c_{k,N}$'s. The coefficients vanish at and beyond both ends of the tower, and

$$\sup_{k \in \mathbb{Z}} |c_{k+1,N} - c_{k,N}| = \frac{1}{N}.$$

This proves (6). $\square$

Proof of Theorem 1. Lemma 4 shows that $S^* - I$ is not bounded below. By (2), $S - I$ is therefore not surjective. If S were generalized hyperbolic, Lemma 2 would say that $I - S$ is surjective, a contradiction. The diffeomorphism h has Axiom A and strong transversality by Lemma 3, and $p = 1$ is included in the universal claim of Conjecture 1. $\square$

4 Remarks on scope and novelty

The counterexample uses the interaction between the derivative cocycle and the volume change in an L^1 norm. It does not decide the conjectured conclusion for volume-preserving Axiom-A maps, for C^0 vector fields, or for other fixed values of p under additional bunching assumptions.

Weighted operators generated by Morse–Smale maps have an established spectral theory [2], and generalized hyperbolicity has been characterized for significant classes of L^p operators [3]. A bounded search on 21 July 2026 found no statement of the local Jacobian-neutral tower obstruction above. This is a novelty check, not a definitive certification.

References

- [1] S. Tikhomirov, *Generalized (C, λ) -structure for nonlinear diffeomorphisms of Banach spaces*, arXiv:2510.05499 (2025), <https://arxiv.org/abs/2510.05499>.
- [2] A. Antonevich and J. Makowska, *Operators generated by Morse–Smale mappings*, arXiv:1303.1293 (2013), <https://arxiv.org/abs/1303.1293>.
- [3] E. D’Aniello, U. B. Darji and M. Maiuriello, *Generalized hyperbolicity and shadowing in L^p spaces*, arXiv:2009.11526 (2020), <https://arxiv.org/abs/2009.11526>.